Write a python program to recognize the number 0, 1, 2, 3, 9. A 5 * 3 matrix forms the numbers. For any valid point it is taken as 1 and invalid point it is taken as 0. The net has to be trained to recognize all the numbers and when the test data is given, the network has to recognize the particular numbers.

In [1]:
```
import numpy as np
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

In [2]:
```
digit_0=[
    [1,1,1],
    [1,0,1],
    [1,0,1],
    [1,0,1],
    [1,1,1]
]

digit_1=[
    [0,1,0],
    [1,1,0],
    [0,1,0],
    [0,1,0],
    [1,1,1]
]

digit_2=[
    [1,1,1],
    [0,0,1],
    [1,1,1],
    [1,0,0],
    [1,1,1]
]

digit_3=[
    [1,1,1],
    [0,0,1],
    [1,1,1],
    [0,0,1],
    [1,1,1]
]

digit_4=[
    [1,0,1],
    [1,0,1],
    [1,1,1],
    [0,0,1],
    [0,0,1]
]

digit_5=[
    [1,1,1],
```

```
    [1,0,0],
    [1,1,1],
    [0,0,1],
    [1,1,1]
]

digit_6=[
    [1,1,1],
    [1,0,0],
    [1,1,1],
    [1,0,1],
    [1,1,1]
]

digit_7=[
    [1,1,1],
    [0,0,1],
    [0,0,1],
    [0,0,1],
    [0,0,1]
]

digit_8=[
    [1,1,1],
    [1,0,1],
    [1,1,1],
    [1,0,1],
    [1,1,1]
]

digit_9=[
    [1,1,1],
    [1,0,1],
    [1,1,1],
    [0,0,1],
    [1,1,1]
]
```

In [3]:
```
x=np.array([
    digit_0,digit_1,digit_2,digit_3,digit_4,digit_5,digit_6,digit_7,digit
])
x=x.reshape(10,15)
print("Training Data Shape: ",x.shape)
```

```
Training Data Shape:  (10, 15)
```

In [6]:
```
y=np.array([0,1,2,3,4,5,6,7,8,9])
model=MLPClassifier(
    hidden_layer_sizes=(15,),
    activation='relu',
    learning_rate_init=0.01,
    solver='lbfgs',
    max_iter=100,
    random_state=42
```

```
)

model.fit(x,y)
print("Training Done")
```

```
Training Done
```

In [8]:
```
pred=model.predict(x)
print("Actual Digit:",y)
print("Predicted Digit: ",pred)
```

```
Actual Digit: [0 1 2 3 4 5 6 7 8 9]
Predicted Digit:  [0 1 2 3 4 5 6 7 8 9]
```

In [9]:
```
acc=accuracy_score(y,pred)
print("Accuracy Score: ",acc*100,"%")
```

```
Accuracy Score:  100.0 %
```

In [11…
```
new_input=[
    [1,1,1],
    [0,0,1],
    [1,1,1],
    [0,0,1],
    [1,1,1]
]
z=np.array(new_input)
z=z.reshape(1,15)

prediction=model.predict(z)
print("New Input: ")
print(np.array(new_input).reshape(5,3))
print("\nRecognised Digit: ",prediction[0])
```

```
New Input:
[[1 1 1]
 [0 0 1]
 [1 1 1]
 [0 0 1]
 [1 1 1]]

Recognised Digit:  3
```